

Qualification Report on 10Gbps EA integrated DFB LD Modules, NX8560 series

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1. Introduction

This report describes the qualification test results on EA (Electro-Absorption Modulator) integrated DFB Laser modules, NX8560 series, for 10Gbps applications.

NEC Compound Semiconductor Devices Ltd.' qualification program is based on the reliability requirements described in [†]Telcordia GR-468-CORE.

NX8560 series EA integrated DFB Laser modules have TEC(Thermo Electric Cooler) elements in their modules, therefore these devices are qualified in accordance with CO (Central Office) environment requirements.

[†] Telcordia is a trademark of Telcordia Technologies, Inc.

2. Module level qualification

2-1. Summary of module qualification

Table-1 shows the summary of our qualification tests on EA integrated DFB Laser modules. Endurance test groups are still on-going, but all the mechanical tests were completed and qualified.

Table 1. Test results on Laser modules qualification

Group	Test	Conditions	SS	C	QTY Fail	Status
Mechanical Integrity	Mechanical Shock	MIL-STD-883E, Condition B $4.9 \times 10^3 \text{ m/s}^2$, 1msec half sine 5 times/axis	11	0	0	Pass
	Vibration	MIL-STD-883E, Condition A $1.96 \times 10^2 \text{ m/s}^2$, 20 – 2000Hz, 4 min/cycle, 4cycles/axis				
	Thermal Shock	$\Delta T = 100^\circ \text{C}$	11	0	0	Pass
	Solderability	No steam aging	11	0	0	Pass
	Fiber Pull	9.8N, 5sec, 3times	11	0	0	Pass
Endurance	Accelerated Aging	Tc=75°C, VMOD was terminated, T _{LD} =25°C, Constant Im APC to keep initial Iop of 80mA	10	0	0	Pass
	High Temp. Storage	Ta=85°C, 2000hrs	11	0	0	On-going
	Low Temp. Storage	Ta=-40°C, 2000hrs	11	0	0	Pass
	Temperature cycling	TH=+85°C, TL=-40°C, 100cyc for Re, 500cyc for info	11	0	0	Pass
	Damp Heat	50°C, 85%RH, 3500 hrs	11	0	0	On-going
Special Tests	Internal Moisture	MIL-STD-883E method 1018	13	0	0	Pass
	ESD threshold	R=1.5k Ω , C=100pF, Vth $\geq$ 500V	**12;EA-LD **14;Moni-PD	0	0	Pass

Notes

*Re; Requirement / Info; Information

** Total 12 devices of EA-LDs and 14 devices of monitor-PDs were tested.

2-2. Mechanical integrity

Table-2 shows mechanical integrity test results.

Table-2. List of Mechanical integrity tests

Test	Conditions	SS	C	QTY Fail	Status
Mechanical Shock	MIL-STD-883E method 2002 Condition B $4.9 \times 10^3 \text{m/s}^2$, 1msec half sine, 5 times/axis	11	0	0	Pass Fig. 1
Vibration	MIL-STD-883E method 2007 Condition A $1.96 \times 10^2 \text{m/s}^2$, 20 – 2000Hz, 4 min/cycle, 4cycles/axis				
Thermal Shock	MIL-STD-883E method 1011 $\Delta T=100^\circ\text{C}$	11	0	0	Pass
Solderability	MIL-STD-883E method 2003 No steam aging	11	0	0	Pass
Fiber Pull	9.8N, 5sec, 3times	11	0	0	Pass Fig. 2

Table 3. Failure criteria

Tests	Parameters (Symbol)	Conditions	Failure criteria
a. Mechanical shock through Vibration b. Fiber Pull	I _{th}	T _{LD} =25°C	**Greater than $\pm 1\text{mA}$ change from the initial value
	η	T _{LD} =25°C, Set I _m the initial (BOL) value to achieve I _{op} of 80mA	Greater than $\pm 0.5\text{dB}$ change from the initial value
	* λ	T _{LD} =25°C, Set I _m the initial (BOL) value to achieve I _{op} of 80mA	Greater than $\pm 100\text{pm}$ change from the initial value
	*R _{TEC} (Resistance of TEC)	T _a =25°C	Greater than $\pm 5\%$ change from the initial value
Thermal shock	-	Examined at X 20 magnification	Any evidence of damage in mechanical integrity.
Solderability	-	Examined at X 10 magnification	At least 95% covered by continuous new solder coating.

* Applicable to Mechanical shock through Vibration tests.

** Due to the small I_{th} value, less than 10mA.

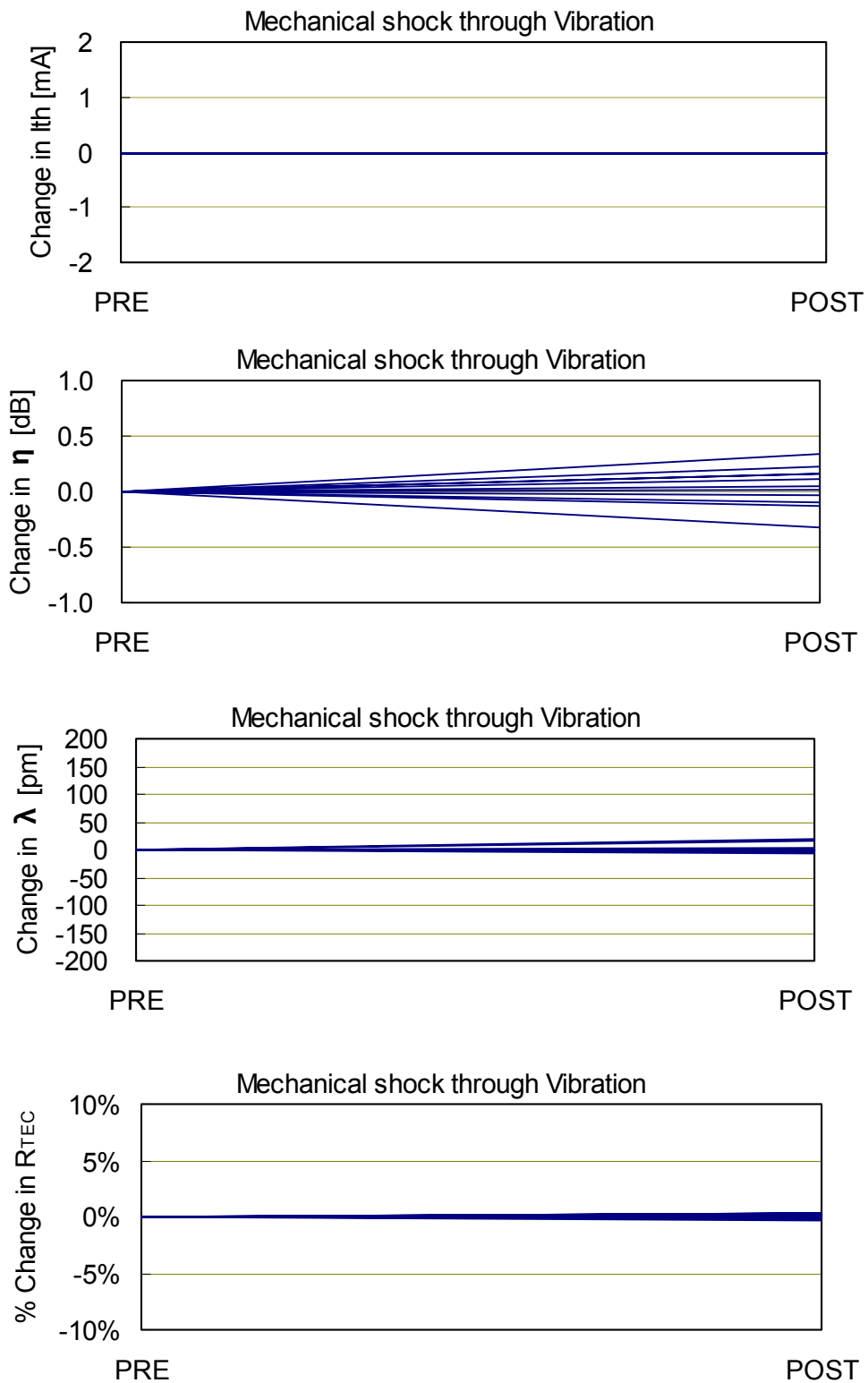


Figure 1. Mechanical shock and Vibration test

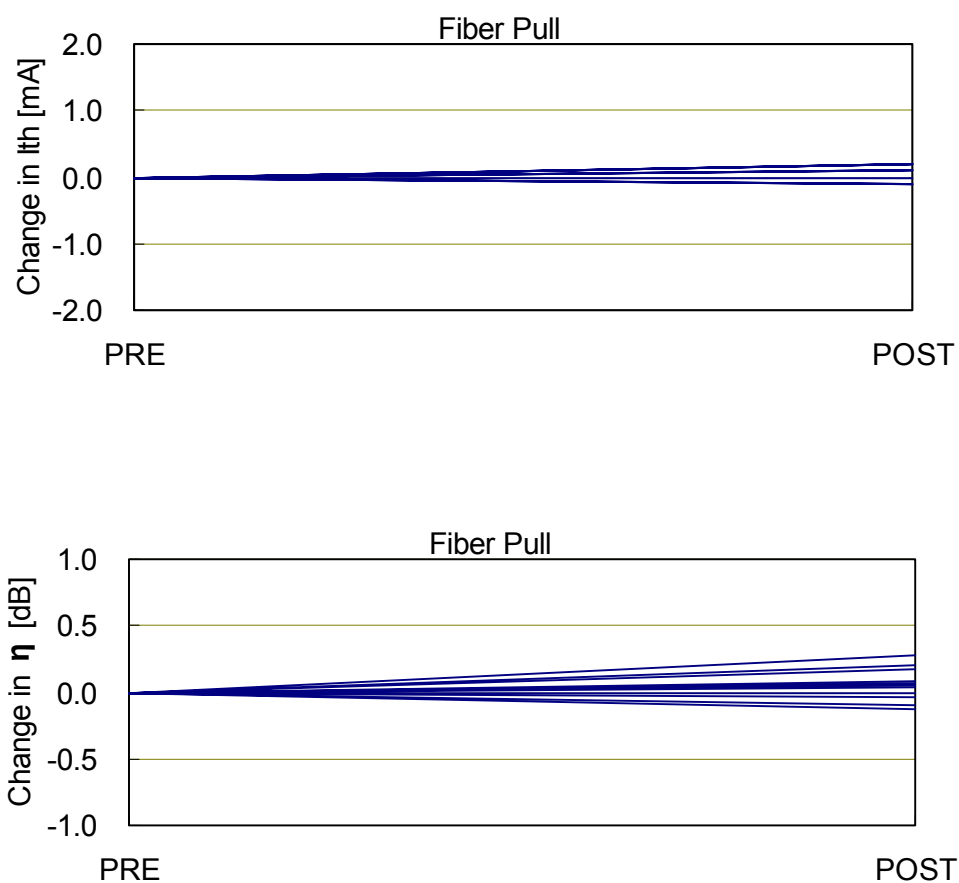


Figure 2. Fiber pull test

2-3. Endurance

Table-4 shows endurance test results. Except for high temperature storage and damp heat tests, other tests are completed and passed the qualification.

Wavelength of peak emission is monitored for the information purpose.

Table 4. List of Endurance tests

Test	Conditions	SS	C	QTY Fail	Status
Accelerated Aging	Tc=75°C, VMOD was terminated, T _{LD} =25°C, Constant Im APC to keep initial lop of 80mA	10	0	0	Pass Fig. 3
High Temp. Storage	Ta=85°C, 2000hrs, 400hrs measurement was completed.	11	0	0	On going Fig. 4
Low Temp. Storage	Ta=-40°C, 2000hrs	11	0	0	Pass Fig. 5
Temperature cycling	TH=+85°C, TL=-40°C, 100cyc for Requirement, 500cyc for information purpose	11	0	0	Pass Fig. 6
Damp Heat	50°C, 85%RH, 3500 hrs, biased 2500hrs measurement was completed.	11	0	0	On going Fig. 7

Table 5. Failure criteria

Tests	Parameters (Symbol)	Conditions	Failure criteria
Accelerated aging	*Pf	Tc=75°C, VMOD was terminated, T _{LD} =25°C,	Greater than ± 0.5 dB change from the initial value
	*lop	Constant Im APC to keep initial lop of 80mA	Greater than $\pm 20\%$ change from the initial value
a. High temp. storage b. Low temp. storage	lth	T _{LD} =25°C	Greater than ± 1 mA change from the initial value
c. Temperature cycling d. Damp heat	η	T _{LD} =25°C, Set Im the initial (BOL) value to achieve lop of 80mA	Greater than ± 0.5 dB change from the initial value
	λ	T _{LD} =25°C, Set Im the initial (BOL) value to achieve lop of 80mA	Greater than ± 100 pm change from the initial value

* As for Accelerated aging, these parameters were monitored at the aging temperature(Tc=75°C, T_{LD}=25°C).

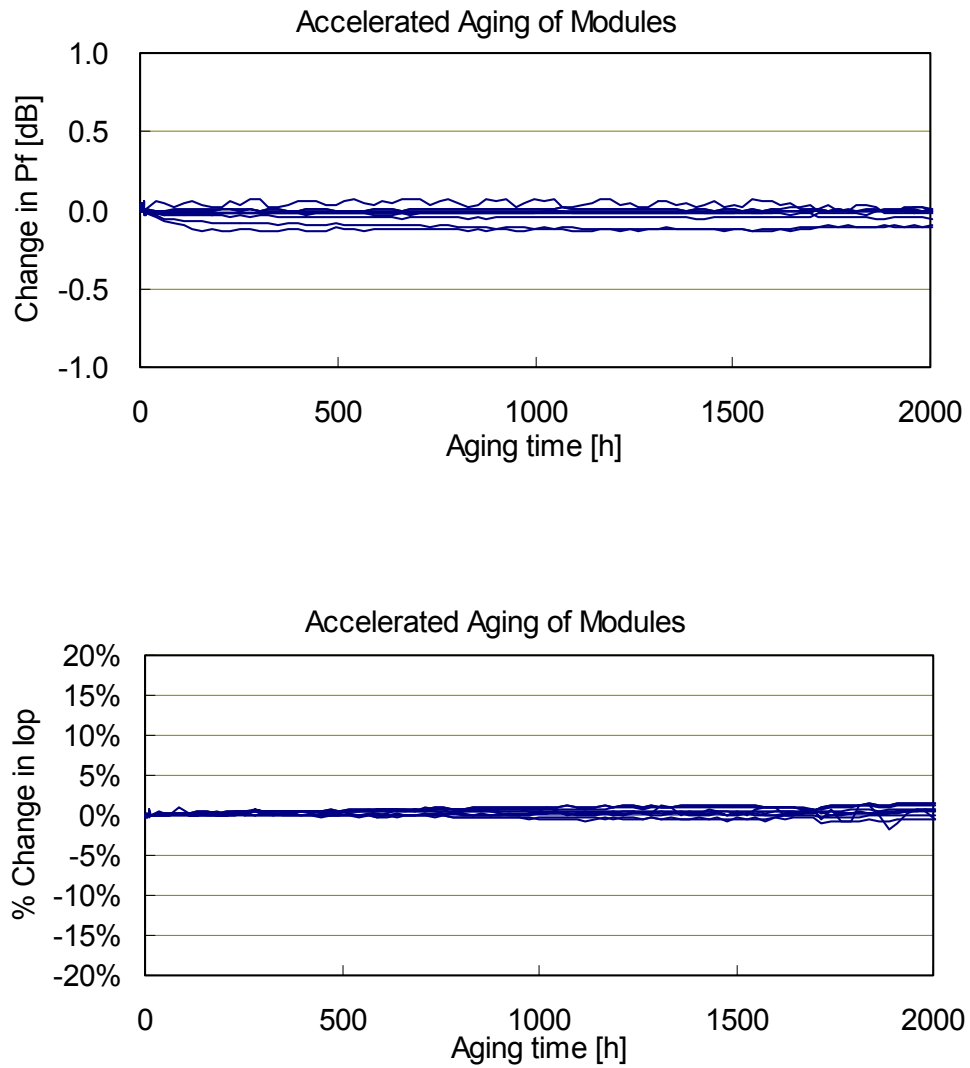


Figure 3. Accelerated aging test

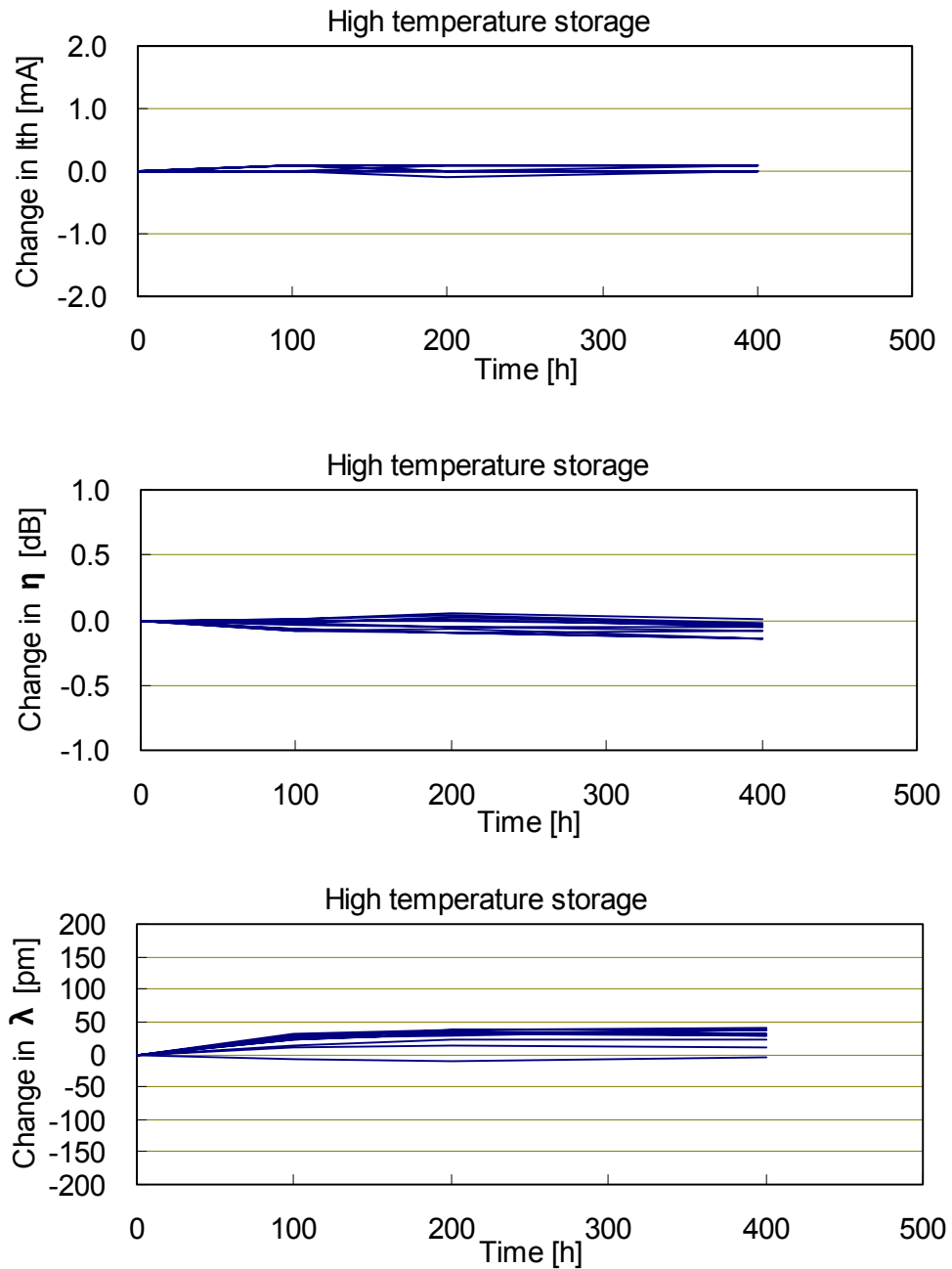


Figure 4. High temperature storage test

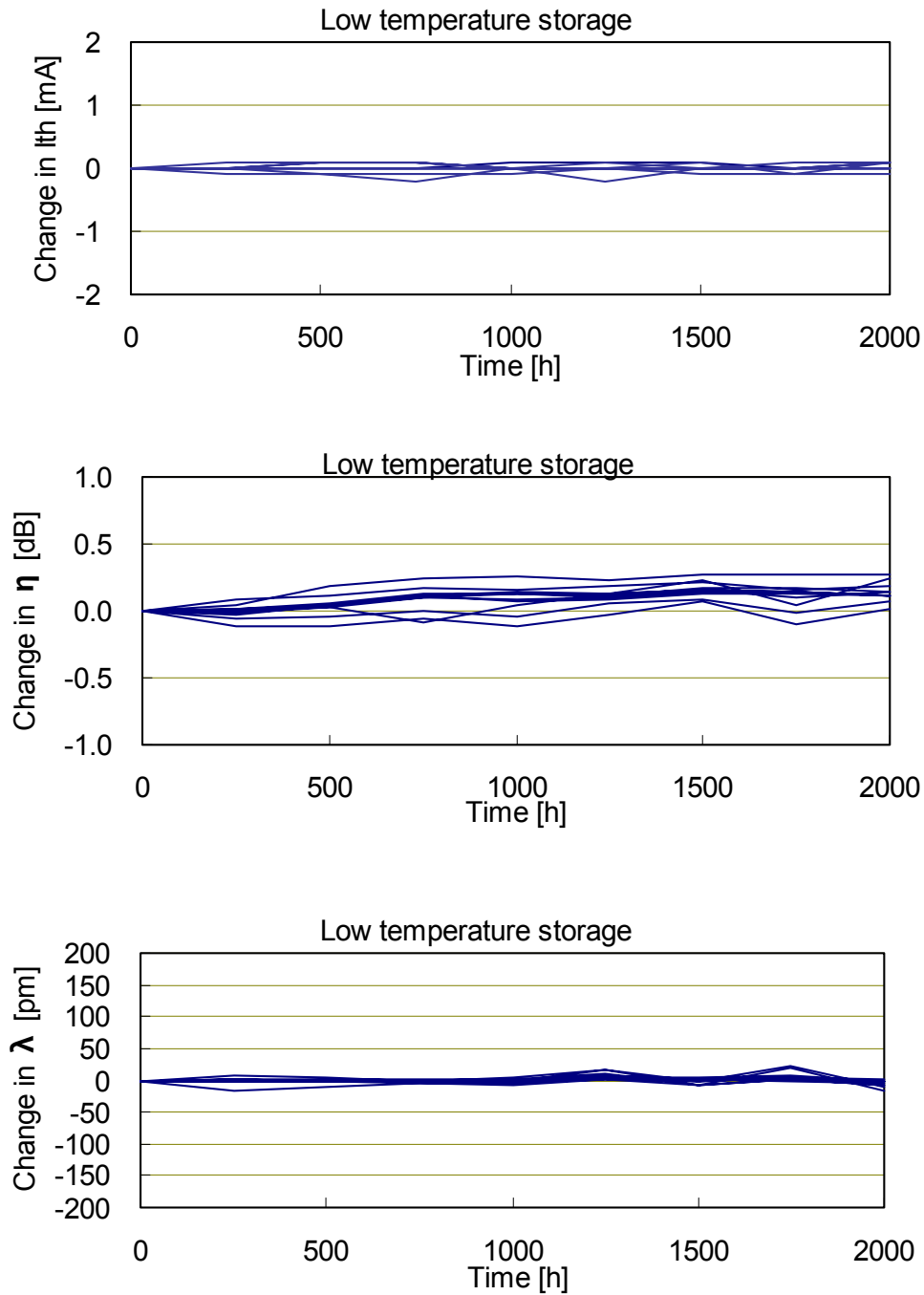


Figure 5. Low temperature storage test

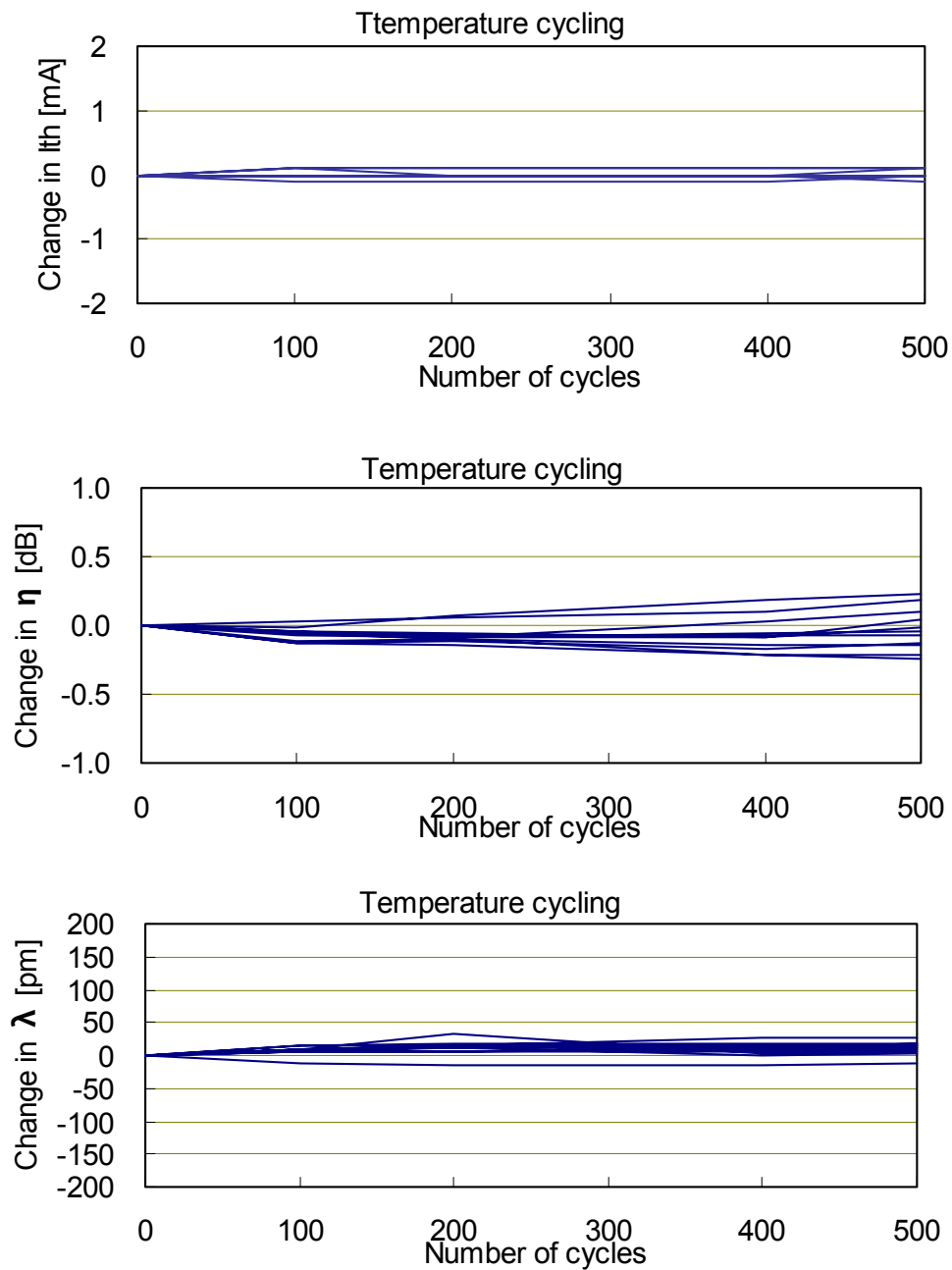


Figure 6. Temperature cycling test

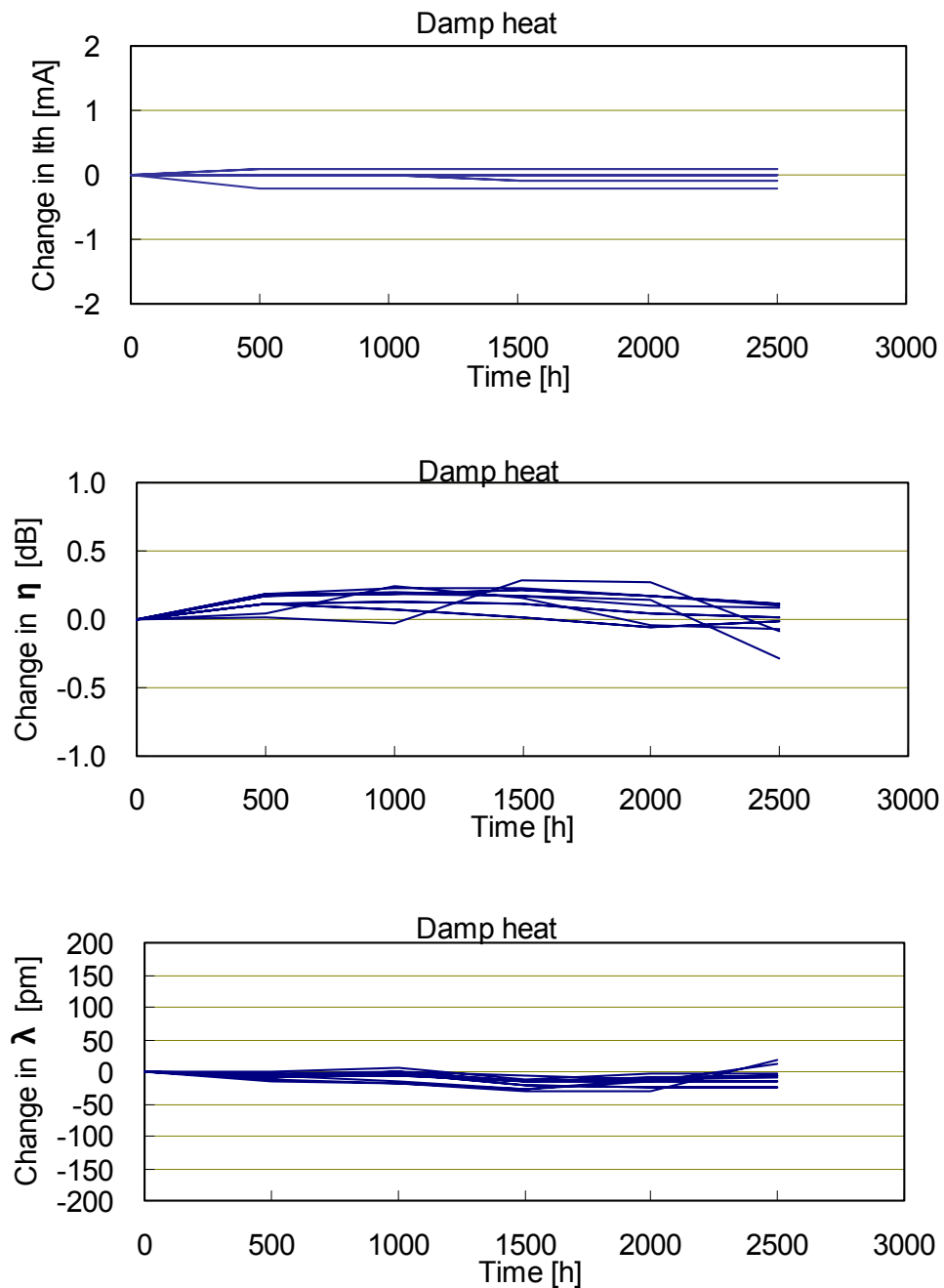


Figure 7. Damp heat test

2-4. Special tests

Table-6 shows special tests results. The most weak part against ESD in LD modules is a monitor PD chip. Both EA section of LDs and monitor PDs showed short phenomenon in electrical characteristics. Minimum ESD threshold of this module stands within 900V to 1000V for Human Body Model.

Table 6. List of Special tests

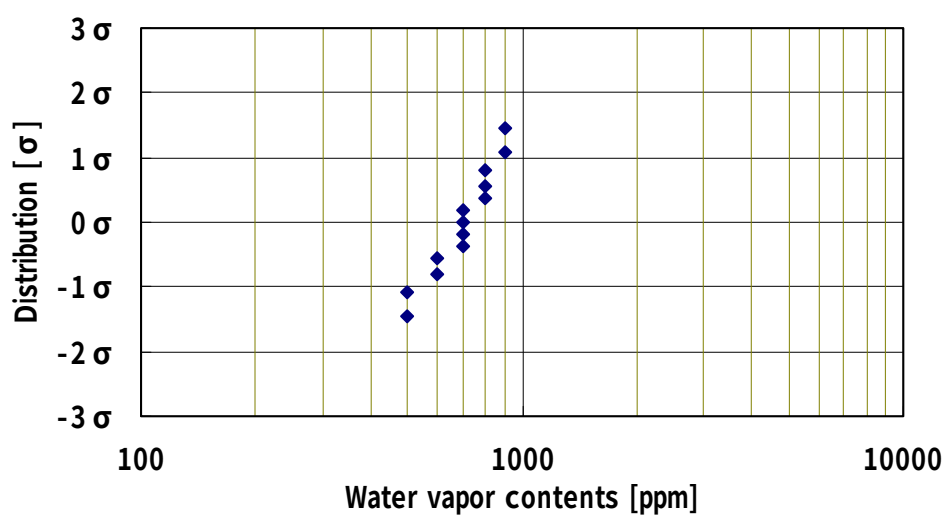
Test	Conditions		SS	C	QTY Fail	Status
Internal Moisture	MIL-STD-883E Method 1018		13	0	0	Pass Fig.8
ESD threshold	MIL-STD-883E method 3015 R=1.5K Ω , C=100pF, Vth $\geq$ 500V	LD of EA-LD	*12	0	0	Pass Fig.9 & Fig.10
		Modulator	*12	0	0	
		Monitor PD	*14	0	0	

* Each sample was divided evenly for POSITIVE (anode-cathode) polarity and NEGATIVE (cathode-anode) polarity pulse tests.

Table 7. Failure criteria

Tests	Parameters	Conditions	Failure criteria
Internal Moisture	Water vapor	-	Greater than 5000ppm water vapor contents
ESD threshold for LD-section of EA-LDs	Ith	Ta=25°C, CW	Greater than ± 1 mA change from the initial value
	Pf	If=80mA Ta=25°C	Greater than ± 0.5 dB change from the initial value
	Vf	Ta=25°C, If=100uA	Greater than $\pm 10\%$ change from the initial value
	IR	Ta=25°C, VR=2V	Greater than +100% change from the initial value
ESD threshold for MOD-section of EA-LDs	*Pf	If _{MOD} =80mA Ta=25°C	Greater than ± 0.5 dB change from the initial value
	Vf	Ta=25°C, If=10mA	Greater than $\pm 10\%$ change from the initial value
ESD threshold for Monitor PDs	Id	Ta=25°C, VR=5V	Greater than 10nA
	VBR	Ta=25°C, IR=100uA	Greater than $\pm 10\%$ change from the initial value

* Modulator shows spontaneous emission when forward biasing, so we checked the fiber coupled power (Pf) to detect the degradation.



The entire specimen showed less than 1000ppm internal water vapor contents.

Figure 8. Internal moisture

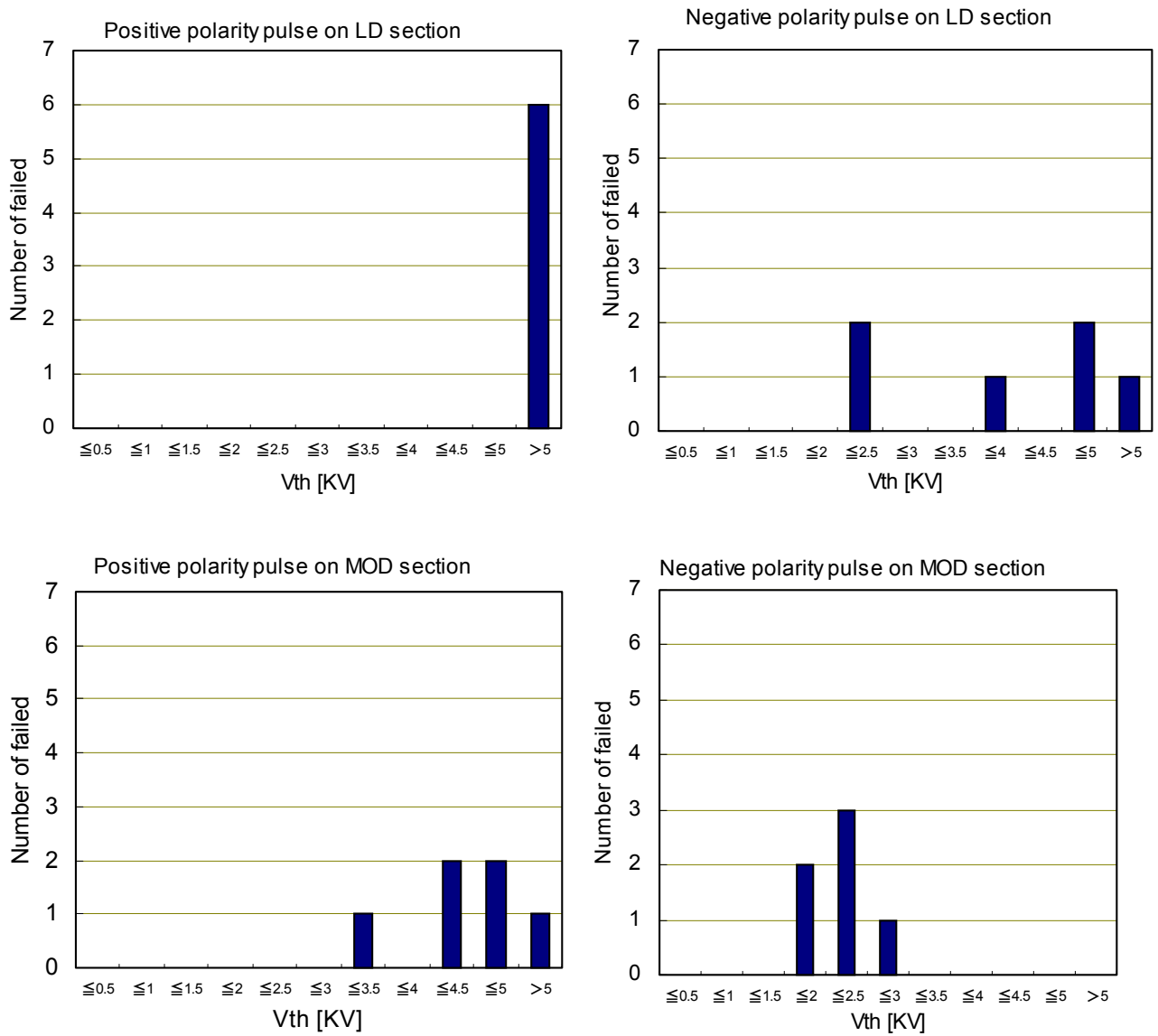


Figure 9-1. Distribution of ESD threshold on EA -LDs

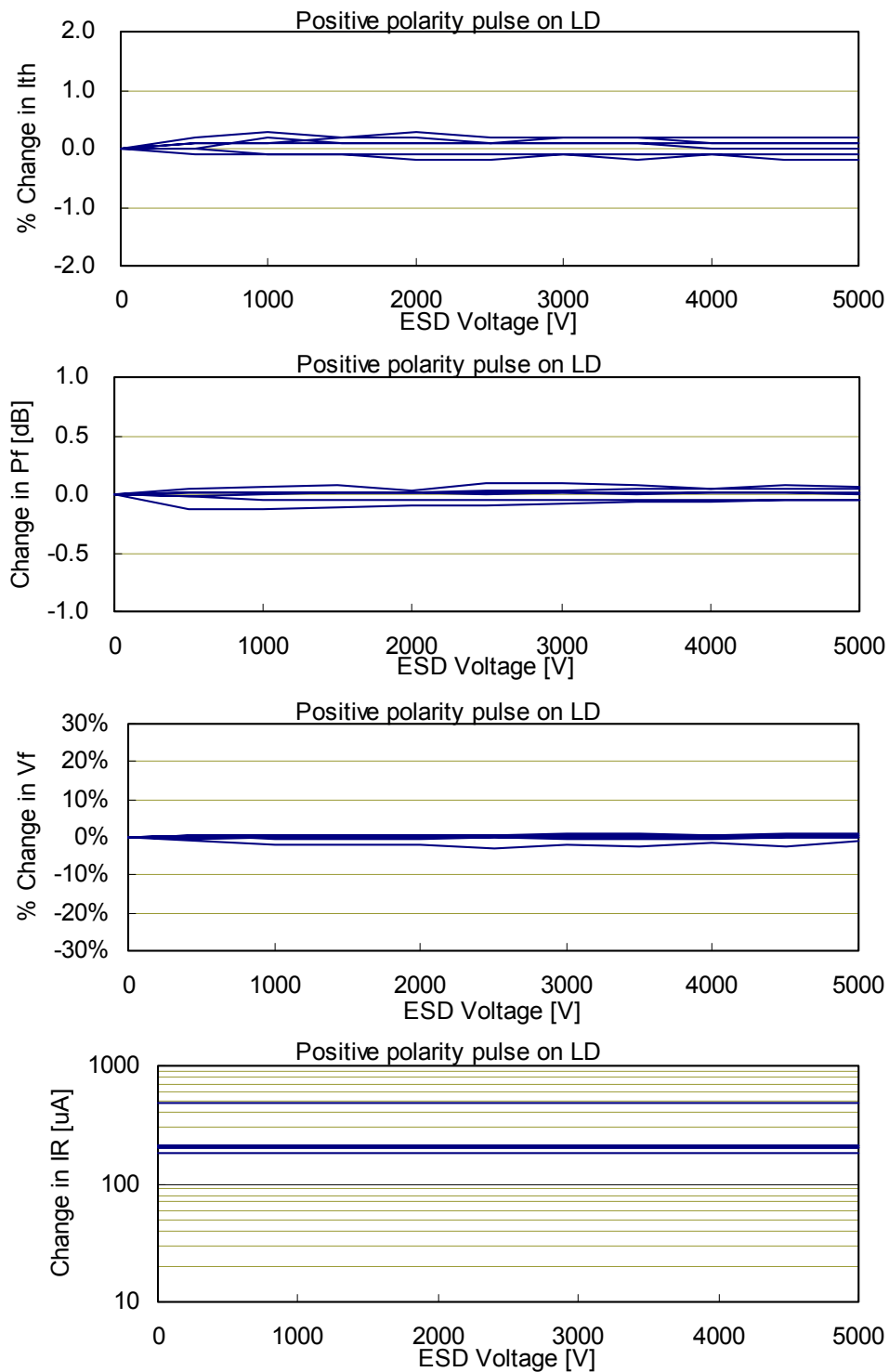


Figure 9-2. Positive Pulse test on LD-section of EA-LDs

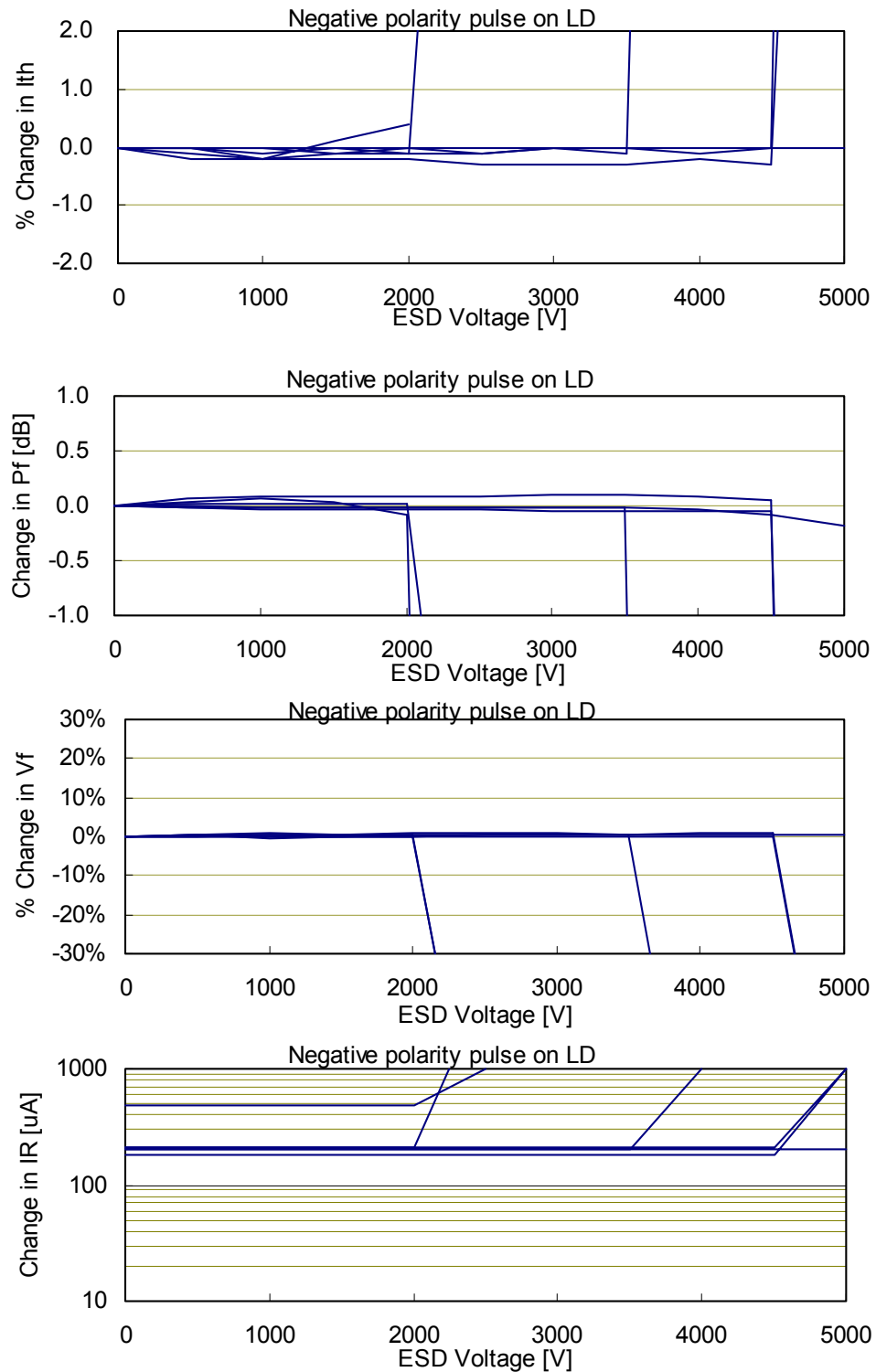


Figure 9-3. Negative Pulse test on LD-section of EA-LDs

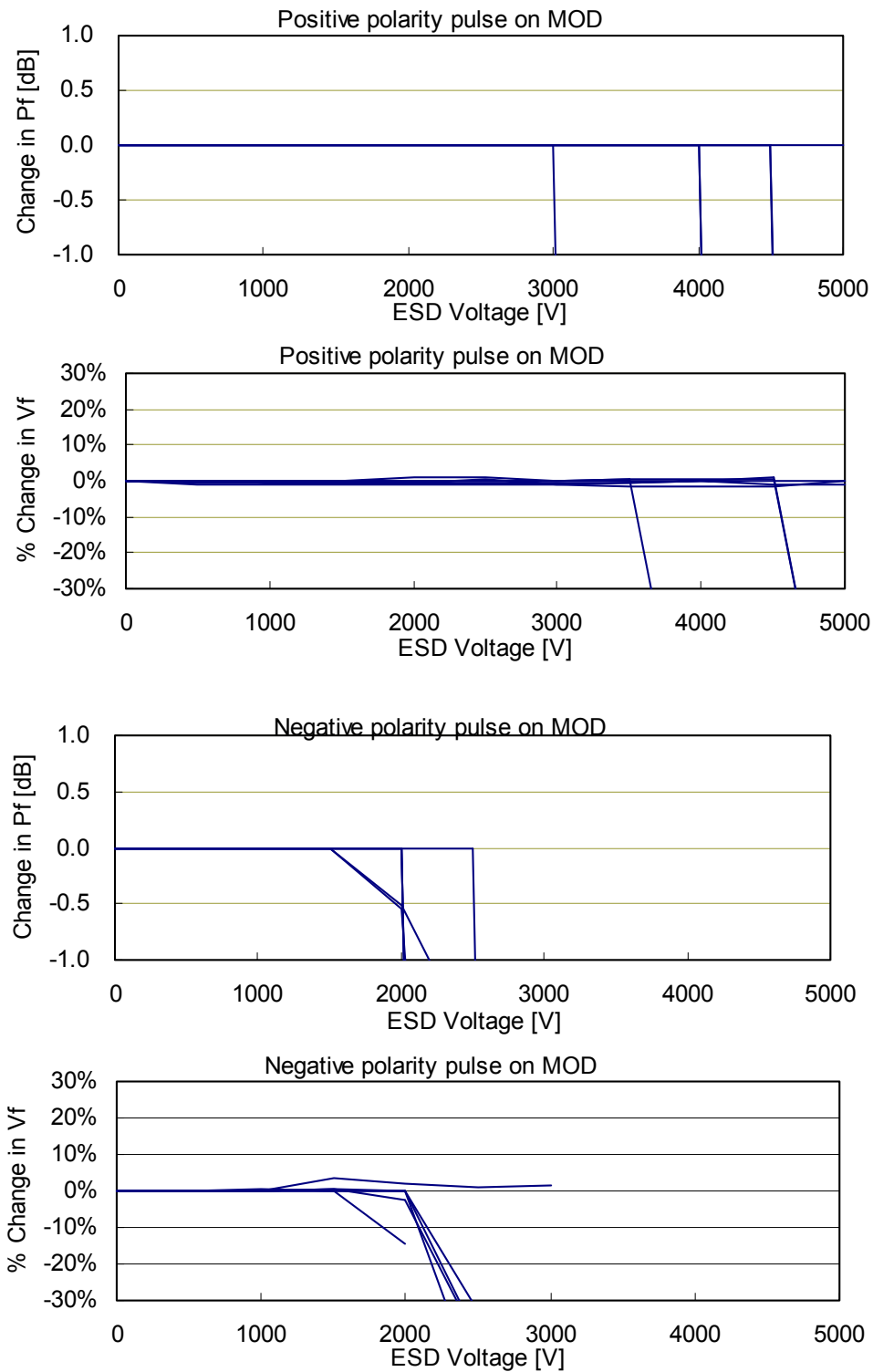


Figure 9-4. Positive or Negative Pulse tests on MOD-section of EA-LDs

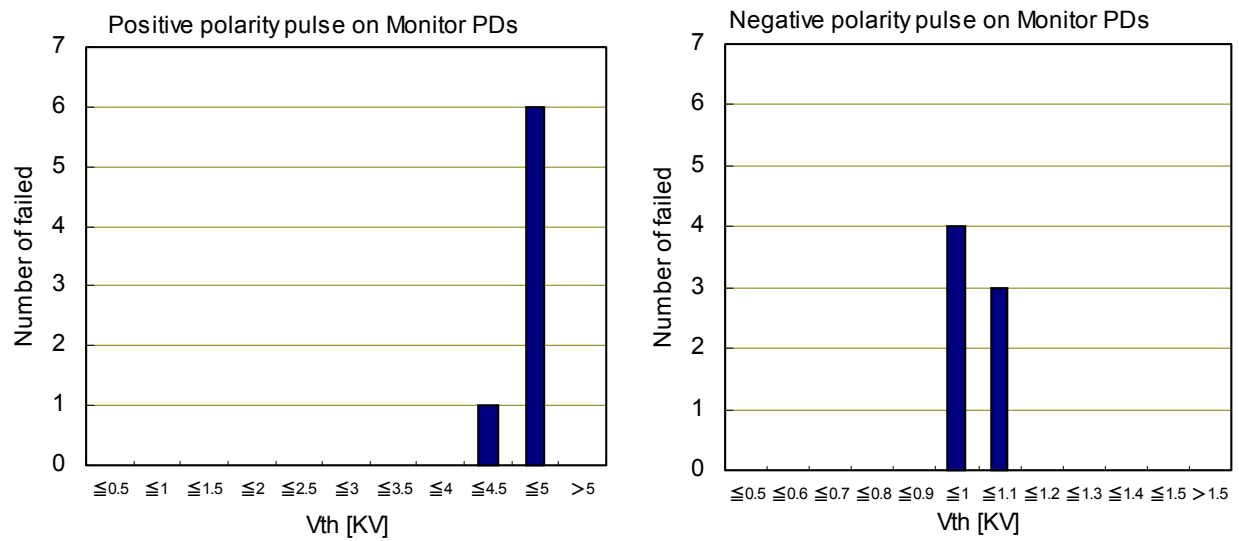


Figure 10-1. Distribution of ESD threshold on Monitor-PDs

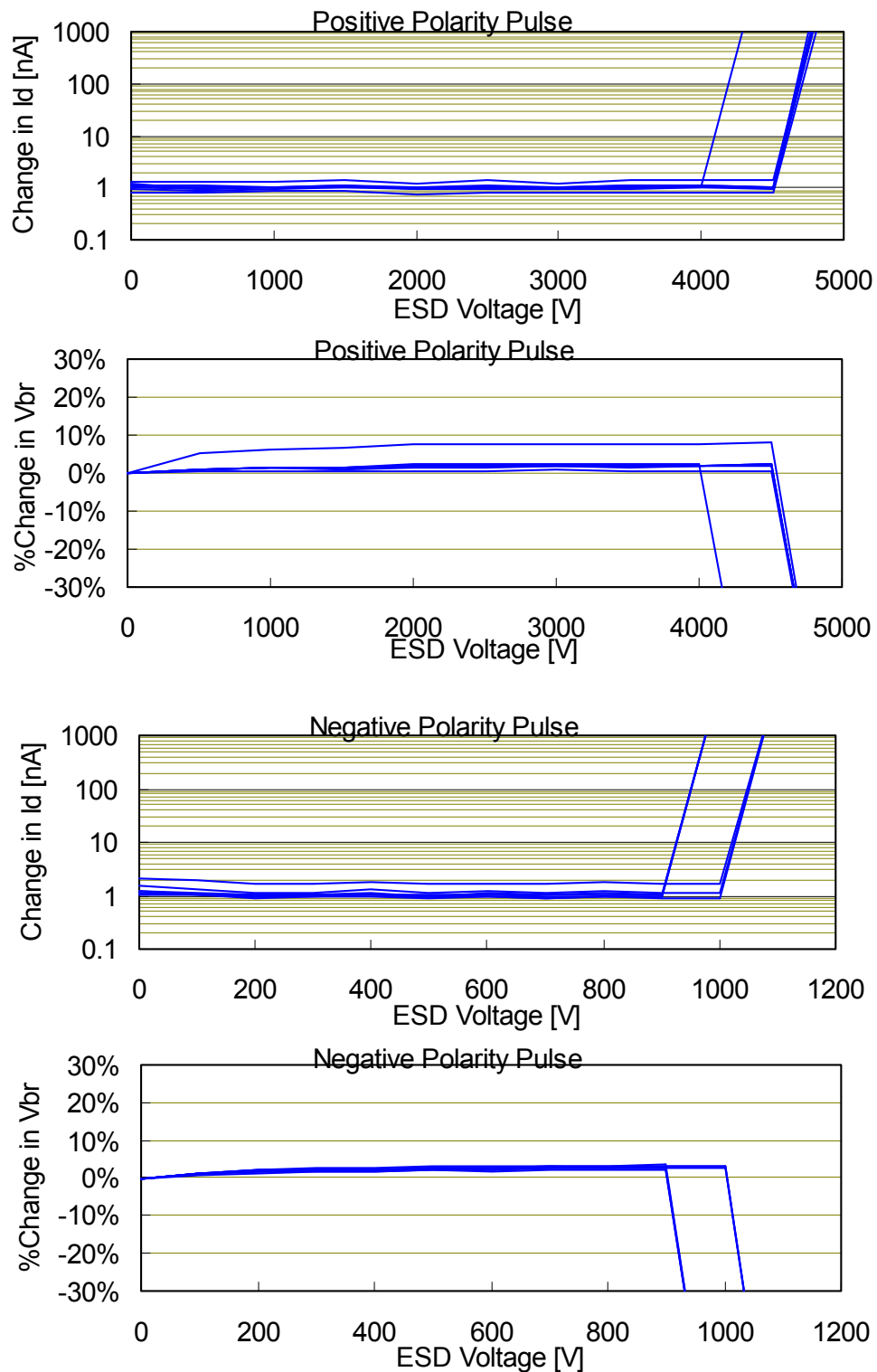


Figure 10-2. Positive or Negative Pulse tests on Monitor-PDs

3. Accelerated aging on EA-LD sub-modules(Laser diodes)

A laser diode shows wear-out degradation. Table-8 shows the accelerated aging test results under APC condition with two different temperatures to demonstrate the long-term operation and to confirm the median life and activation energy.

Table 8. Accelerated aging test results on LD sub-modules

Test	Reference Document	Conditions	SS	C	QTY Fail	Status
Accelerated Aging	† Telcordia GR-468-CORE	T _{LD} =60 °C, V _{MOD} =-3V, 4000hrs, APC with rear facet monitor current at **80% of EA-LD maximum power	18	-	*0	Fig.11
		T _{LD} =75 °C, V _{MOD} =-3V, 4500hrs, APC with rear facet monitor current at **80% of EA-LD maximum power	12	-	*0	Fig.12

* No sudden failure was observed.

** Power level of 80mA drive current (I_{op}) at T_{LD}=25°C is about 80% of EA-LD maximum power, so accelerated aging of LD was performed at 80% of maximum power.

Table 9. End of life criteria for LD sub-modules

Parameters (Symbol)	Conditions	EOL criteria
I _{op}	T _{LD} = Aging temperature	Greater than +20% change from the initial value

† Telcordia is a trademark of Telcordia Technologies, Inc.

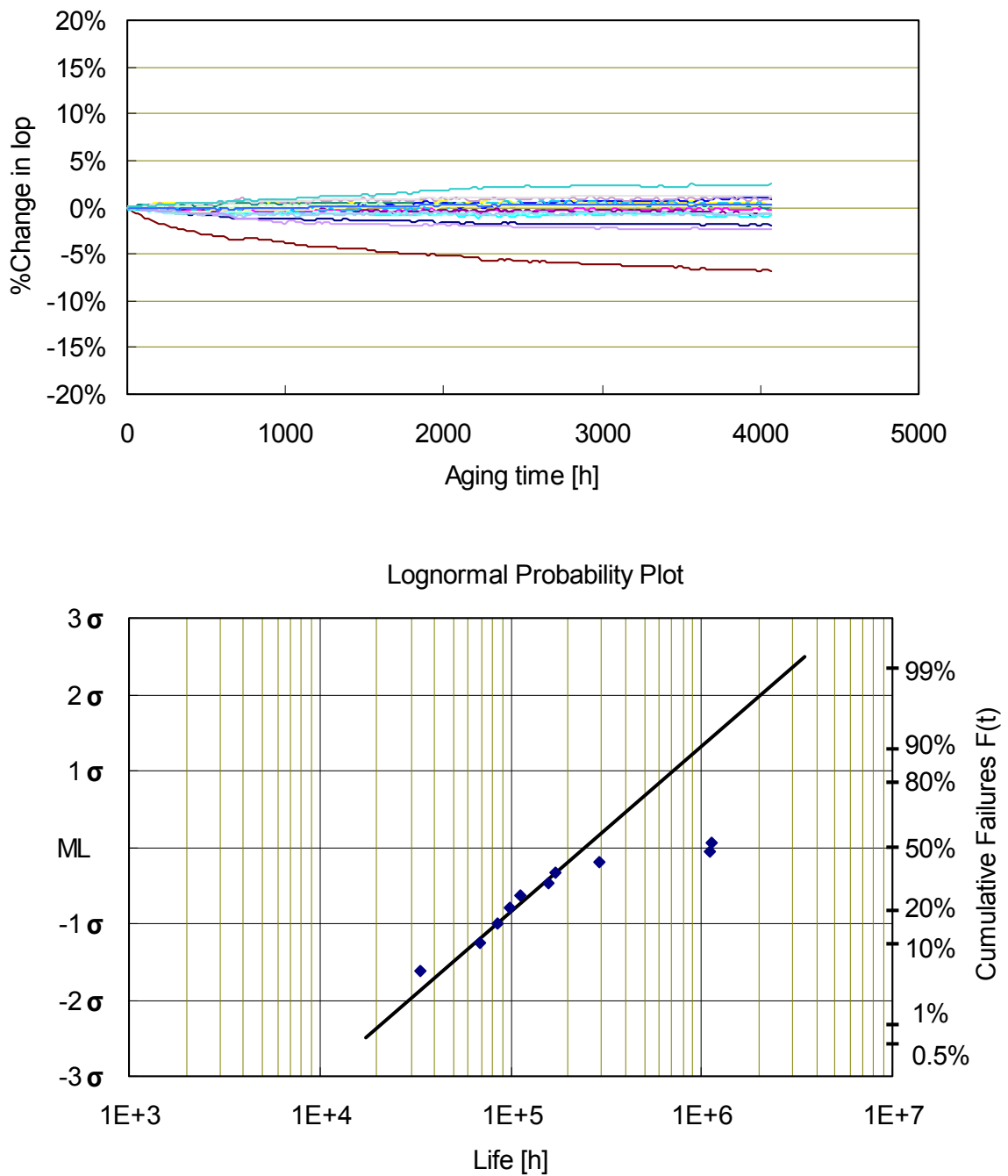


Figure 11. Accelerated aging (APC) test results on LD sub-modules at 60°C

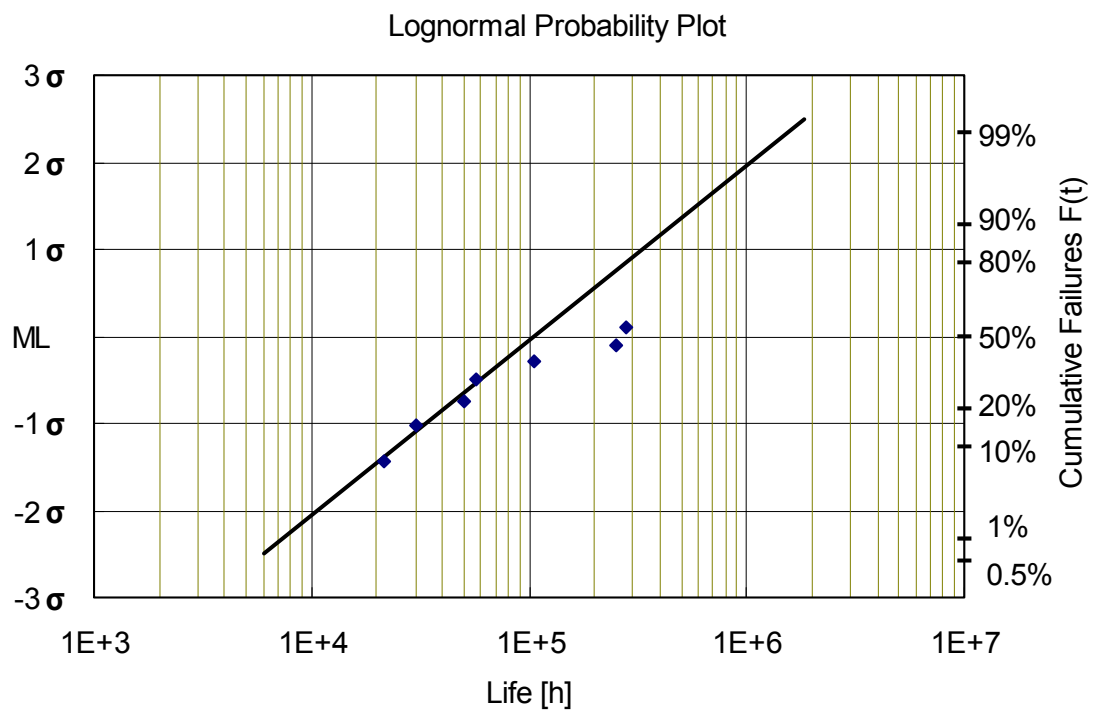
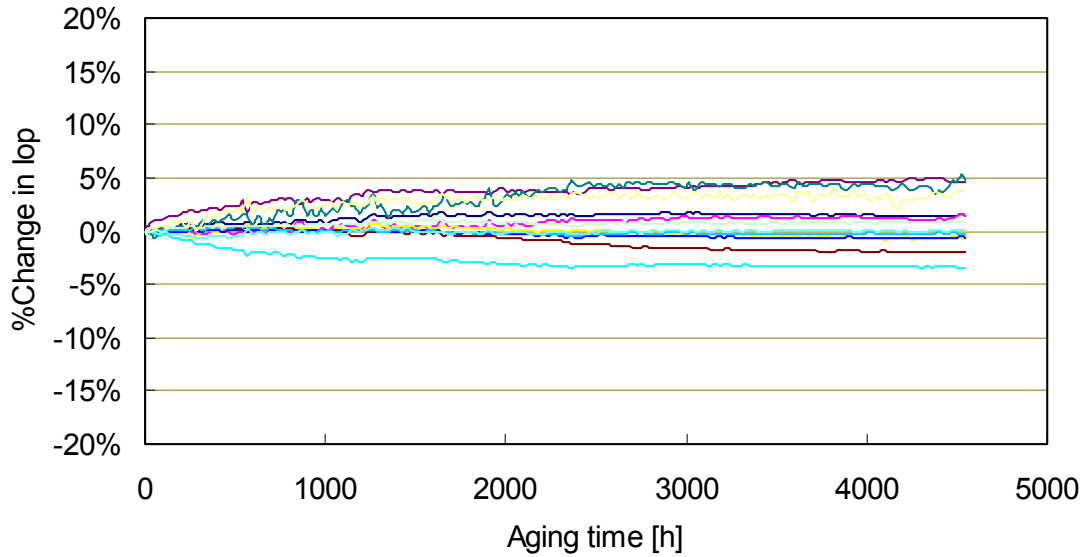


Figure 12. Accelerated aging (APC) test results on LD sub-modules at 75°C

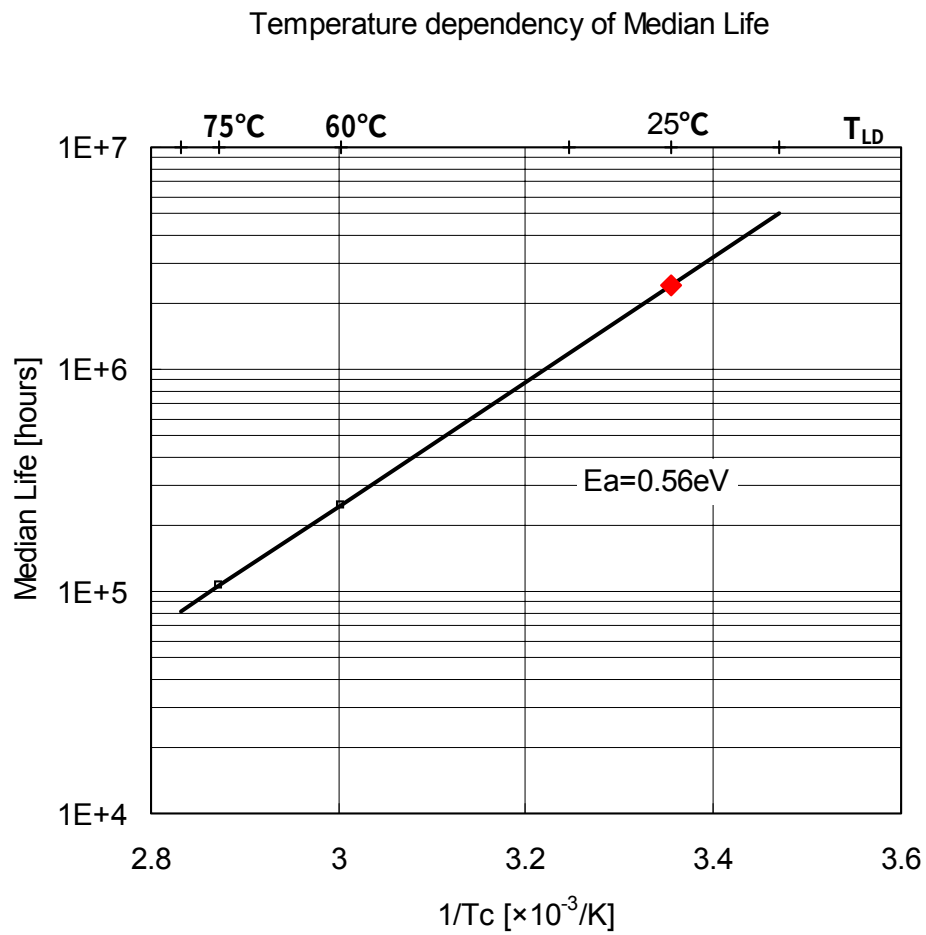


Figure 13. Estimation of Activation Energy of LD sub-modules

4. Life estimation

Table 10. Reliability Parameters

Failure Mode	Item	LD	Other parts
		$T_{LD}=25^{\circ}\text{C}$	
Wear-out failure @ $T_c=45^{\circ}\text{C}$	Ea [eV] (Activation Energy)	0.56 Measured value	No wear-out failure was observed
	σ (Standard deviation)	1.1	
	ML [hrs] (Median Life)	2.4×10^6	
	$\lambda_w(t)$ [Fits] @ 10 years @ 15 years @ 20 years @ 25 years (Failure rate)	45.3 85.6 124.2 158.5	

Failure Mode	Item	LD module
Random failure	λ_R [Fits] (Actual failure rate)	10G EML is now under development phase, there is no actual random failure rate data. But the failure rate of 2.5G EML is expected to be less than 40 Fit (Calculated value is 37.3 Fit at the end of 2001), so λ_R of 10G EML is expected to be low value.

5. Conclusion

The full qualification tests described in [†]Telcordia GR-468-CORE have not been completed yet, but no failure was observed in the previous tests documented in this report. The similar LD module, such as NX857x which is a wavelength locked CW LD module, have already passed the remaining tests of high temperature storage and damp heat tests, so NEC Compound Semiconductor Devices, Ltd. considers the module will be successfully qualified based on CENTRAL OFFICE application requirements in GR-468-CORE.

[†] Telcordia is a trademark of Telcordia Technologies, Inc.

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